A Nontrivial Tauberian Pair in $L_1(0,1)$ Outside $\mathcal{K}_+$

FA/Banach automatic math run

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Source And Target

The source is Manuel Gonzalez and Antonio Martinez-Abejon, *Tauberian pairs of closed subspaces of a Banach space*, arXiv:2603.27754. For closed subspaces M, N of a Banach space Z, the source associates the pair (M, N) with the operator

$$Q_N J_M : M \longrightarrow Z/N.$$

The pair is tauberian when $Q_N J_M \in \mathcal{W}_+$, and it belongs to $\mathcal{K}_+$ when $Q_N J_M$ is upper semi-Fredholm.

The final questions in the source ask, among other things, for non-trivial examples in $Z = L_1(0,1)$ of tauberian pairs

$$(M, N) \in \mathcal{W}_+ \setminus \mathcal{K}_+.$$

The source notes that (M, M), with M an infinite-dimensional reflexive subspace of $L_1(0,1)$, is a trivial example. The construction below gives an explicit example in which both entries M and N are nonreflexive and distinct.

The Construction

Partition $(0,1)$ into three measurable sets A, B, C of positive measure. Then

$$L_1(0,1) = L_1(A) \oplus_1 L_1(B) \oplus_1 L_1(C).$$

Choose:

- an infinite-dimensional reflexive closed subspace $E \subset L_1(A)$, for instance the closed span of the Rademacher functions on A, which is isomorphic to ℓ_2 by Khintchine's inequality;
- a closed subspace $F \subset L_1(B)$ isometric to ℓ_1, for instance the closed span of normalized indicators of a countable measurable partition of B;
- a closed subspace $G \subset L_1(C)$ isometric to ℓ_1, chosen similarly inside C.

Set

$$M := E \oplus_1 F, \qquad N := E \oplus_1 G,$$

as closed subspaces of $Z := L_1(0,1)$.

Theorem 1. *The pair (M, N) is a tauberian pair in $Z = L_1(0,1)$, but $(M, N) \notin \mathcal{K}_+$. Moreover, both M and N are nonreflexive.*

Proof. The spaces M and N are nonreflexive because they contain the closed subspaces $F \cong \ell_1$ and $G \cong \ell_1$, respectively.

We first prove that (M, N) is tauberian. Let

$$P_A,\ P_B,\ P_C : L_1(0,1) \to L_1(0,1)$$

be the band projections onto the three summands. They extend to projections on Z^{**}, and hence induce projections

$$\widehat{P}_A,\ \widehat{P}_B,\ \widehat{P}_C : Z^{\mathrm{co}} := Z^{**}/Z \to Z^{\mathrm{co}}$$

whose sum is the identity.

We use the source characterization: a pair (M, N) is tauberian if and only if

$$M^{\mathrm{co}} \cap N^{\mathrm{co}} = \{0\}$$

inside Z^{co}. Let $\xi \in M^{\mathrm{co}} \cap N^{\mathrm{co}}$. Since $M \subset L_1(A) \oplus_1 L_1(B)$, the C-band component of every element of M^{co} is zero, and therefore

$$\widehat{P}_C \xi = 0.$$

Similarly, since $N \subset L_1(A) \oplus_1 L_1(C)$, the B-band component of every element of N^{co} is zero, so

$$\widehat{P}_B \xi = 0.$$

It remains only to check the A-band component. From $\xi \in M^{\mathrm{co}}$, the element $\widehat{P}_A \xi$ belongs to the image of E^{**} in Z^{co}. But E is reflexive, so $E^{**} = E \subset Z$, and this image is zero in the quotient Z^{**}/Z. Hence

$$\widehat{P}_A \xi = 0.$$

Since the induced band projections sum to the identity on Z^{co}, we get $\xi = 0$. Thus $M^{\mathrm{co}} \cap N^{\mathrm{co}} = \{0\}$, and the pair is tauberian.

Finally,

$$\ker(Q_N J_M) = M \cap N = E,$$

because the B- and C-supports are disjoint. The space E is infinite-dimensional. Therefore $Q_N J_M$ is not upper semi-Fredholm, so $(M, N) \notin \mathcal{K}_+$. $\square$

Remark 1. *This example is deliberately elementary. It does not rely on the probabilistic Johnson–Nasseri–Schechtman–Tkocz construction of a tauberian operator on $L_1(0,1)$ that is not upper semi-Fredholm. The flexibility of pairs is enough: the common reflexive part E makes the kernel infinite-dimensional, while the disjoint nonreflexive ℓ_1 parts keep both entries of the pair nonreflexive without creating an intersection in the quotient Z^{co}.*

Novelty Check

The cheap run indexes were searched for arXiv:2603.27754, “Tauberian pair”, “closed subspaces”, “$L_1(0,1)$”, “upper semi-Fredholm”, and related Tauberian/weak compactness keywords. No prior run packet for this source or for this explicit pair construction was found. Nearby entries concern Tauberian/semi-embedding questions for different papers, or weak compactness questions for unrelated state-space and tensor-product settings.

Human Review Recommendation

Review as a literal full answer to the source's request for non-trivial examples in $L_1(0,1)$. The only substantive verification is the computation $M^{\mathrm{co}} \cap N^{\mathrm{co}} = \{0\}$, which follows from the three disjoint band projections and the reflexivity of E.